cerbyl

-Learning, unified

Your brain is deleting everything you studied...

Scientists proved this in 1885, AI tools still ignore this in 2025

Time after learning	Retention
20 minutes	58%
1 hour	44%
24 hours	33%
1 week	21%
1 month	~10%

The Ebbinghaus Forgetting Curve shows that humans forget roughly 50% of new information within an hour and 90% within a month if it is not reviewed. This rapid, exponential decline in retention demonstrates that traditional, one-time learning events fail, creating a critical need for spaced repetition and active recall tools to solidify long-term memory. [1]

Why chatbots fail at learning?

	AI chatbots	Cerbyl
What it stores	Facts you tell it	Every convo, quiz score, note, flashcard, weak topic, updated to your profile for future context
How it's used	Passive	Active - Drives what content you see next
Spaced repetition	None	Built into quiz and flashcard engine
Concept dependency	None	Ensures you know that X is a prerequisite for Y before learning Y
Acts on your gaps	No - waits for you to ask	Yes - Proactively resurfaces weak topics

"AI tools can remember facts. It won't tell you that you scored 40% on Limits last Tuesday and should review it before attempting Derivatives today. Memory and adaptive learning are not the same thing."

Why cerbyl?

Cerbyl is the first student platform engineered around the forgetting curve.

- Every topic you study is logged to your **personal episodic memory** (ChromaDB vector store)
- Your **knowledge graph** (Neo4j) tracks mastery level per concept and knows your prerequisites
- When you open Cerbyl next week, the AI already knows you forgot Mitosis and haven't touched Limits since Tuesday
- Quizzes and flashcards surface weak topics automatically - spaced repetition built into the engine, not bolted on

We don't just answer your question. We remember your answer and come back for it.

Before	After
Quizlet anki®	*“One tool to rule them all”*

The Opportunity

$26B → $133B Global Digital Education Market | 2024 - 2030 | 31.5% CAGR

86%	**74.4%**	**31.5%**
Of students report using AI predominantly for their studies. [2]	Dominance of STEM - Our core student Demographic. [3]	Annual Growth Rate - Faster than SaaS median [3]

Europe's e-learning market is booming!

- Companies building tools for teachers are capturing one side of the classroom.
- But the student side, personalization, adaptive learning, AI-driven practice remains fragmented and underdeveloped.

$26B → $81.5B	**21.1% CAGR**	**Blended Learning**
Europe corporate e-learning market 2024-2030	Fastest growing edtech region after Asia Pacific	#1 fastest growing segment in Europe - exactly what Cerbyl enables

[4]

Why we thrive in this market

- 74.4% of this market is STEM
- Exactly the subject area where concept prerequisites matter most, where "just Google it" fails, and where Cerbyl's knowledge graph (Neo4j concept dependency mapping) has the deepest structural advantage.
- A student learning calculus without mastering limits needs a system that *knows* that gap exists. Cerbyl does.

Student expectations

01- The study

We surveyed 100+ students in 2025.

The average student uses **4.3 apps** to cover a single study workflow - AI, notes, flashcards, quizzes, calendar. None of them talk to each other. Context dies at every switch.

02 - Expectation gap

Gen Z grew up with Social media knowing their next video before they do and Spotify building a playlist around their mood.

The most personalized experience in a student's day currently is their entertainment app, not their education.

03 - Missed opportunity

AI tools exist, students use them. However, ChatGPT does not know about the quiz you scheduled on google calendar for next week to help you improve nor does Anki tailor your flashcards to your notes you took on notion.

04. The cerbyl standard

Every action in Cerbyl is an input to an AI actively modeling your understanding, not just answering your questions.

4.3 apps. Zero shared context. We built the one.

Three systems. One brain.

1. The Learning Graph (Neo4j)

A real-time knowledge graph that maps every concept a student has studied, their mastery level, prerequisite gaps, and common mistakes. When you ask for a flashcard on "derivatives," Cerbyl already knows you're weak on "limits"

2. Episodic Memory (ChromaDB)

Every note taken, flashcard reviewed, quiz answered, and chat session is stored as a semantic vector in your personal memory store. The AI retrieves relevant past sessions to personalize every new interaction like a tutor who actually remembers you.

3. The Adaptive Engine (LangGraph)

A multi-node AI pipeline (not a single prompt) that: fetches your context ,builds a personalized prompt and generates calibrated content. Difficulty, depth, and tone adapt per student, per session, per topic.

One note. Seven intelligent actions.

1. Student creates a note on "Mitosis"
2. Cerbyl writes an episode to ChromaDB (note_activity event)
3. Neo4j updates concept node: Mitosis → linked to Cell Division, DNA Replication
4. Student then opens Quiz Hub → LangGraph fetches: weak topics from episodic memory + prerequisite gaps from Neo4j
5. Quiz is generated at appropriate difficulty calibrated to your weak topics
6. Score < 60% → Neo4j records update_struggle on Mitosis
7. Next flashcard session → Mitosis surfaces first via spaced repetition

Duolingo made language learning addictive

Cerbyl does the same for every subject..

- Every action earns XP create a note (20pts), ace a quiz (30pts), master a flashcard set (10pts), study an hour (50pts). Harder work, more points. Every session compounds.
- Daily streak milestones for weekly and monthly progress along with achievement milestones across each activity. Showing up consistently is recognised, tracked and celebrated
- Duolingo gamifies repetition. Cerbyl gamifies mastery. Streaks reward showing up but underneath it all, the knowledge graph tracks whether you actually learned, not just whether you logged in.

Cerbyl is a social platform. Not just a study tool.

01- Friends

Share your Notes, Flashcards, AI chat sessions etc directly with friends, with permission controls (view or edit).

Track their learning activity to see who is more locked in this semester and hold each other accountable.

Compare stats, streaks and share study material with them!

02 - Learning playlists

Curate collections of study content - like Spotify playlists but for knowledge.

Make them public for the whole community or collaborative so friends can add to them together.

03. Quiz battles

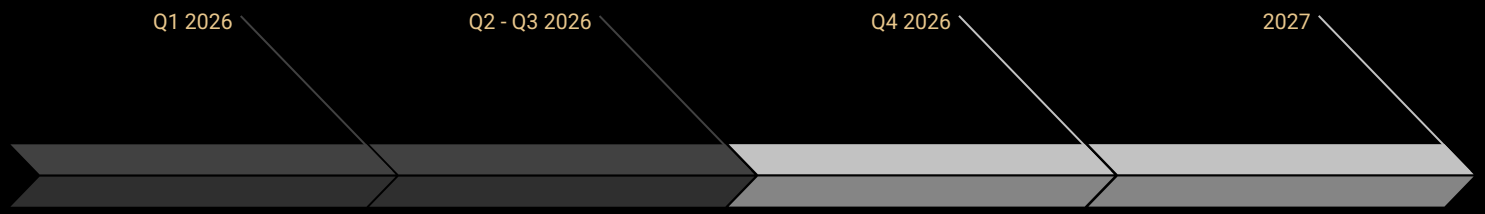
Challenge a friend to a real-time quiz battle on any topic.

Competitive learning is proven to increase engagement and retention.

The Magic

- One subscription replaces multiple tools
- No app switching
- Clear progress tracking is proven to fuel motivation
- Every interaction updates your learning profile
- Feels like a personal tutor, not a chatbot

The Roadmap



Phase 1

- AI-powered notes, flashcards, quizzes, and tutor - fully connected
- Episodic memory, knowledge graph, adaptive difficulty engine
- Gamification and social layer

Phase 2

- Native mobile app (iOS + Android) - learning on the go
- School admin panel: district-wide deployment, usage analytics, curriculum upload

Phase 3

- Cerbyl partnerships with American & European institutions
- Multi-language platform support (French, German, Spanish priority)

Phase 4

Open API for builders. Schools, EdTech companies, and developers get direct API access to Cerbyl's adaptive engine

The Team

Aditya Lanka

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"We are not studying this problem from the outside.
We are the users.
We built this because we lived the frustration."

References

[1] https://en.wikipedia.org/wiki/Forgetting_curve

[2] https://www.grandviewresearch.com/industry-analysis/digital-education-market-report

[3] https://www.marketsandmarkets.com/Market-Reports/digital-education-market-7751828.html

[4] https://www.grandviewresearch.com/horizon/outlook/corporate-e-learning-market/europe